

Effect of Continuous Transition between Force Stimuli on Haptic Force Discrimination Tasks

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- Haptic perception is a fundamental sensory ability that enables us to engage with and understand the physical world through touch.
- In this work, we focused on **kinesthetic force perception**.
- Similar to other kinds of stimuli, human perceptual limitations for force stimuli are quantified by the **Just Noticeable Difference (JND)** or **Weber fraction** (δ)¹
 - **JND** — smallest perceivable change in the reference force (F_r).
 - **Weber fraction** — smallest perceivable relative change in the reference force (F_r).

¹E. H. Weber et al., 1996

Conventional Force Discrimination

- Conventional haptic force discrimination tasks involve perceptual differentiation between stimulus pairs separated by a brief inter-stimulus interval (ISI)².

- Weber Fraction (δ):

$$\delta = \frac{|F_t - F_r|}{F_r} \quad (1)$$

- F_r = reference force, F_t = test force

- Perceptual Deadband Region:

$$[(1 - \delta)F_r, (1 + \delta)F_r] \quad (2)$$

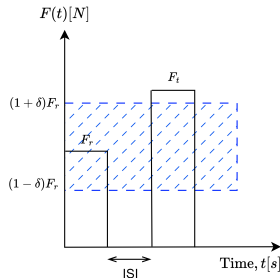


Fig. 1. Force Discrimination with ISI

²L. A. Jones et al., 2012

Objective

- It is unclear if the force JND/Weber fraction remains consistent for **Continuous Force Perception** when measured with conventional methods (with ISI).

- Examples:

- Haptic robot teleoperation³
- Advanced prosthetic limbs⁴
- Virtual reality⁵

- In this study:** We determine how the continuous transition between force stimuli affects force perception.

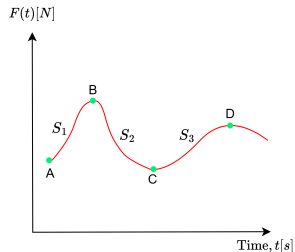


Fig. 2. Example of continuous force transition

³ R. V. Patel et al., 2022 ⁴ R. E. Fan et al., 2008 ⁵ Y. Shi et al., 2024

A. Experimental Setup

- **Device:**

PHANToM Premium 1.5 HF haptic device used for force feedback.

- **Stimulus:**

One-dimensional (1D) haptic force stimulus applied along the positive Z-axis.



Fig. 3. Experimental setup with PHANToM device

B. Force Stimuli

- Perceptual experiment involved two different transition pattern conditions:
 - a Time-separated transition
 - b Continuous linear transition

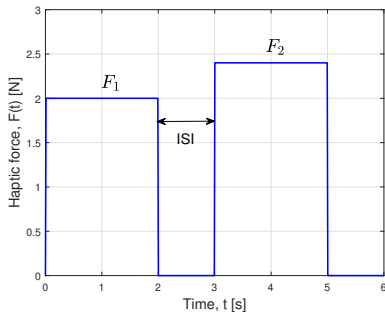


Fig. 4(a). Time-separated transition

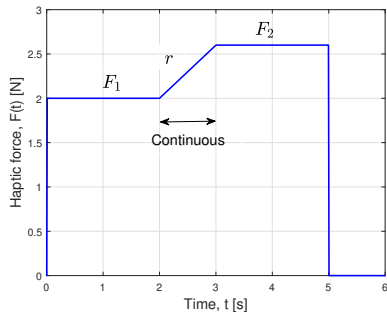


Fig. 4(b). Continuous linear transition

- One of the forces, F_1 or F_2 , is randomly chosen as the reference force ($F_r = 2$ N), and the other as the comparison force (F_c).
- Force levels:
 - a **Time-separated condition:**
 $F_{c1} = \{1.2, 1.4, 1.6, 1.8, 2.0, 2.2, 2.4, 2.6, 2.8\}$ N (9 levels)
 - b **Continuous linear condition:**
 $F_{c2} = \{1.0, 1.2, 1.4, 1.6, 1.8, 2.0, 2.2, 2.4, 2.6, 2.8, 3.0\}$ N (11 levels)

C. Procedure

- Classical Psychophysical Method:
 - Method of Constant Stimuli
 - Randomized stimulus order
- 2-Alternative Forced Choice (2-AFC) paradigm
- Each comparison force (F_c) presented 50 times in a pseudo-randomized order

D. Data Collection

- Participants: 10 users (8M + 2F)
- Trials:
 - 450 trials ($9 F_c \times 50$) – time-separated
 - 550 trials ($11 F_c \times 50$) – continuous linear
- Trial duration: ~ 5 –8 sec
- Total per user: ~ 2.5 hours

Psychometric Functions: by fitting Cumulative Gaussian Functions³

- $JND = P_{84} - P_{50}, \delta = \frac{JND}{F_r}$
- $PSE = P_{50}$
- Slope of psychometric curve

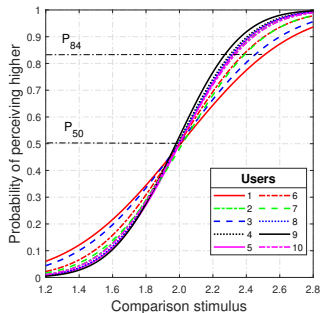


Fig. 5(a). Psychometric fit – Time-separated condition

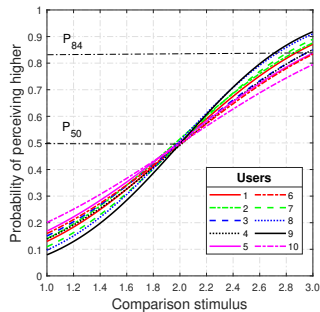


Fig. 5(b). Psychometric fit – Continuous condition

³K. Zchaluk et al., 2009

Results (cont.)

Statistical Results:

- Weber fraction range (across all users):
 - Time-separated: [14.97, 25.87]%
 - Continuous linear: [35.44, 50.55]%
- Paired sample t-test results:
 - WF: $t(9) = -14.80$, $p = 0.000$
 - Slope: $t(9) = 11.94$, $p = 0.000$
 - PSE: $t(9) = -0.23$, $p = 0.824$

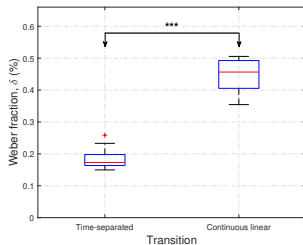


Fig. 6. Comparison of Weber fraction across conditions

Interpretation:

- Higher mean WF in continuous (44.63%) vs. time-separated (18.64%)
- Suggests harder discrimination for continuous transitions
- Lower slope in continuous linear $\rightarrow$ reduced perceptual sensitivity

Continuous Force Perception in Haptic Data Compression

- A perceptually adaptive sampling method (based on Weber's law) is used in haptic data compression for teleoperation.
- Only perceptually relevant points (outside the deadband) are transmitted (red dots, Fig. 7a).
- Signal reconstruction at the receiver typically uses sample-and-hold interpolation (black-dashed, Fig. 7a), resulting in sharp transitions (zero ISI).
- However, to reflect continuous force perception, gradual transitions (e.g., linear interpolation—blue dashed in Fig. 7b) are required.
- Continuous perception enables more data compression since the Weber fraction is higher than the time-separated condition.

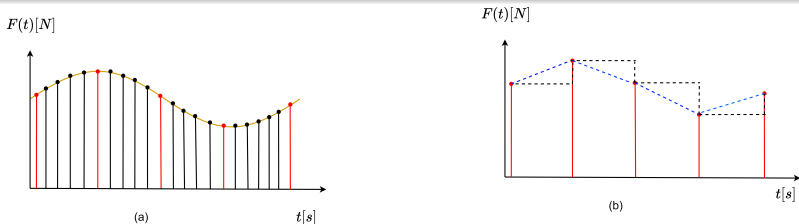


Fig. 7(a). Transmitter side (brown color)

Fig. 7(b). Receiver side

Thank You!